Journal of Electronics, Electromedical Engineering, and Medical Informatics
Homepage: jeeemi.org; Vol. 8, No. 3, July 2026, pp: 330-340 e-ISSN: 2656-8632

RESEARCH ARTICLE OPEN ACCESS

Modelo Predictivo de Riesgo de Desnutrición Crónica Infantil en el Perú mediante Machine Learning

Abel Is-Boset Guevara Huasco[1], Zulin Pamela Vallejos Cotrina[2], and Verónica Lisseth Vergara Rojas[3]

[1] Escuela Profesional de Ingeniería de Sistemas, Facultad de Ingeniería y Arquitectura, Universidad Peruana Unión, Lima, Perú

Corresponding author: Abel Is-Boset Guevara Huasco (e-mail: abelguevara@upeu.edu.pe), **Author(s) Email**: Zulin Pamela Vallejos Cotrina (e-mail: zulin.vallejos@upeu.edu.pe), Verónica Lisseth Vergara Rojas (e-mail: veronica.vergara@upeu.edu.pe)

Abstract (Arial 10, Bold) ….

Keywords …. (Bold, Arial 10)

I. Introduction

(ALL Arial 10) La desnutrición crónica infantil constituye uno de los principales desafíos de salud pública a nivel mundial, ya que afecta el desarrollo físico, cognitivo y social de millones de niños menores de cinco años [1][2]. Este problema no solo compromete el bienestar y la calidad de vida de la infancia, sino que también limita el progreso económico y social de las naciones, dificultando el cumplimiento del Objetivo de Desarrollo Sostenible 2: Hambre Cero [3]. La desnutrición infantil es un fenómeno complejo, influido por múltiples factores interrelacionados, como las condiciones económicas, la educación materna, el acceso a servicios básicos, la calidad del entorno ambiental y las desigualdades territoriales [4][5]. En el caso del Perú, la desnutrición crónica infantil continúa siendo un problema estructural: según la Encuesta Demográfica y de Salud Familiar (ENDES, 2023), la prevalencia nacional alcanza el 11,5 %, con regiones como Huancavelica y Loreto que superan el 20 % [15]. Esta desigualdad evidencia que el problema responde también a factores socioeconómicos, sanitarios y ambientales que varían de forma significativa según el territorio.

Este estudio se justifica desde tres ámbitos complementarios. Desde el ámbito teórico, contribuye al conocimiento existente al incorporar los determinantes sociales, ambientales y espaciales en el análisis de la desnutrición infantil mediante enfoques de machine learning, superando las limitaciones de los modelos estadísticos tradicionales centrados únicamente en variables antropométricas [17][18]. Desde el punto de vista práctico, la propuesta busca desarrollar un sistema de pronóstico geoespacial que permita identificar zonas con mayor probabilidad de riesgo nutricional, de modo que instituciones como el Ministerio de Salud, el MIDIS y los gobiernos regionales puedan priorizar intervenciones preventivas y optimizar la distribución de recursos [16][19][20]. Desde el enfoque innovador, el estudio propone la integración de modelos híbridos de machine learning con análisis geoespacial, un enfoque poco explorado en el contexto peruano y latinoamericano, que ofrece además un marco metodológico replicable para otras problemáticas sociales [21][22].

Durante la última década, diversas investigaciones han empleado métodos estadísticos y de machine learning para analizar la desnutrición infantil, logrando avances significativos en la clasificación del estado nutricional y la identificación de factores asociados. Modelos híbridos y de ensamble, como los propuestos por Alam et al. y Hadikurniawati et al., alcanzaron altos niveles de precisión al integrar variables antropométricas, demográficas y socioeconómicas [8][9]. De forma complementaria, estudios con enfoque espacial, como los de Habtewold et al. y Egbon et al., incorporaron análisis bayesianos y geográficos para identificar zonas críticas y patrones territoriales del riesgo nutricional [10][11]. En África Subsahariana, Backer y Billing desarrollaron un modelo basado en Random Forest para predecir la prevalencia de desnutrición aguda considerando variables ambientales y de conflicto [26], mientras que Tadesse et al. aplicaron un enfoque de Gradient Boosting combinado con señales satelitales, alcanzando un AUC de 0.86 en predicciones a seis meses [27]. En el ámbito del análisis espacial, Aheto et al. propusieron un modelo bayesiano espaciotemporal para estimar la desnutrición crónica en Ghana [28], y Mgomezulu et al. combinaron algoritmos de ensamble con técnicas de

Manuscript received 8 April 2025; Revised 10 June 2025; Accepted 15 June 2025; Available online 27 June 2025
Digital Object Identifier (**DOI**): https://doi.org/10.35882/jeeemi.v7i3.948
Copyright © 2026 by the authors. This work is an open-access article and licensed under a Creative Commons Attribution-ShareAlike 4.0 International License (CC BY-SA 4.0).

balanceo de clases (SMOTE), logrando una precisión del 99 % en Malawi [13]. Estas experiencias evidencian el creciente potencial de la inteligencia artificial para mejorar el diagnóstico y el pronóstico de la desnutrición infantil, especialmente en contextos donde la información es limitada o dispersa.

A pesar de estos avances, se identifican limitaciones metodológicas específicas que restringen la aplicabilidad y la generalización de los modelos existentes. La mayoría de estos estudios se ha desarrollado en contextos de Asia y África, centrándose en variables antropométricas individuales sin integrar de manera conjunta las dimensiones sociales, ambientales y territoriales [12][14]. Además, el tratamiento combinado de datos espaciales y socioeconómicos presenta dificultades debido a la heterogeneidad de las fuentes, la diferencia en sus escalas y unidades, y la ausencia de técnicas capaces de vincular de manera coherente la información tabular con la información geográfica [11][13]. Por otro lado, los modelos híbridos que integran machine learning y análisis geoespacial se han explorado escasamente en América Latina, región donde las particularidades geográficas y las desigualdades territoriales exigen enfoques adaptados a su propio contexto [32]. En el caso peruano en particular, los estudios nacionales aún no aplican enfoques predictivos ni análisis geoespaciales que permitan anticipar zonas de riesgo [16], lo que evidencia una brecha entre el conocimiento internacional disponible y su aplicación efectiva al contexto local.

Por lo tanto, el presente estudio tiene como objetivo desarrollar un modelo predictivo de riesgo de desnutrición crónica infantil en el Perú mediante machine learning, que permita analizar territorialmente los factores predominantes por departamento y estimar el riesgo nutricional a nivel individual y territorial.

Las contribuciones de este estudio son las siguientes: 1) la depuración y estructuración de una base de datos consistente que integra el estado nutricional infantil a nivel nacional a partir de los microdatos de la ENDES correspondientes al periodo 2007-2024; 2) la comparación sistemática de algoritmos de machine learning (Regresión Logística, Árbol de Decisión, XGBoost, LightGBM, CatBoost y Red Neuronal) bajo un mismo protocolo de validación cruzada estratificada, con el fin de seleccionar el modelo con el mejor desempeño predictivo; 3) la evaluación del modelo mediante métricas de validación (Accuracy, Precision, Recall, F1-score y AUC) y la interpretación de la influencia de las variables mediante valores SHAP, lo que incrementa su capacidad explicativa; y 4) la implementación de un sistema interactivo de análisis territorial que permite visualizar el riesgo de desnutrición según el perfil ingresado, facilitando su aplicación por parte de instituciones públicas de salud [34][37].

Este artículo se organiza de la siguiente manera: la Sección II describe la metodología empleada, incluyendo el diseño CRISP-DM, la construcción de la base de datos y el modelo híbrido propuesto. La Sección III presenta los resultados obtenidos en la comparación de algoritmos y en la interpretación de variables mediante SHAP. La Sección IV discute la interpretación de los resultados, su comparación con estudios similares y las limitaciones del estudio. Finalmente, la Sección V presenta las conclusiones, replanteando los objetivos, los principales hallazgos y las líneas de trabajo futuro.

II. Method (Heading 1, Arial 10, Bold, Blue)

A. Dataset (Heading 2, Bold, Arial 10,)

(Arial 10, Normal) This study aims to examine whether there is a significant difference in accuracy when the orientation position is different. The three types of orientation used were wrist fully supinated, at rest, and fully pronated, each marked as 1, 2 and 3 orientation as shown in Fig. 1. The EMG signal was recorded using a Bagnoli EMG recorder (Delsys, Massachusetts, USA) with sampling frequency of 4,000 Hz. Afterwards, six pairs of electrodes marked 1 to 6 were placed equidistant on the forearm (Fig. 1). The dataset used in this study was obtained from 10 subjects with normal body condition and non-amputee. The data collection followed ethical clearance procedures as standard of measurement to the human. In these procedures, each subject performed hand gesture movements with three different orientations. In addition, each orientation consists of seven basic movements with each consisting of three levels of contraction (low, medium and high). Furthermore, each level of contraction consisted of three trials. The motion that will be recognized in this study are hand close (C1), hand open (C2), wrist extension (C3), wrist flexion (C4), wrist ulnar deviation (C5), wrist radial deviation (C6), and

Manuscript received 8 April 2025; Revised 10 June 2025; Accepted 15 June 2025; Available online 27 June 2025
Digital Object Identifier (**DOI**): https://doi.org/10.35882/jeeemi.v7i3.948
Copyright © 2026 by the authors. This work is an open-access article and licensed under a Creative Commons Attribution-ShareAlike 4.0 International License (CC BY-SA 4.0).

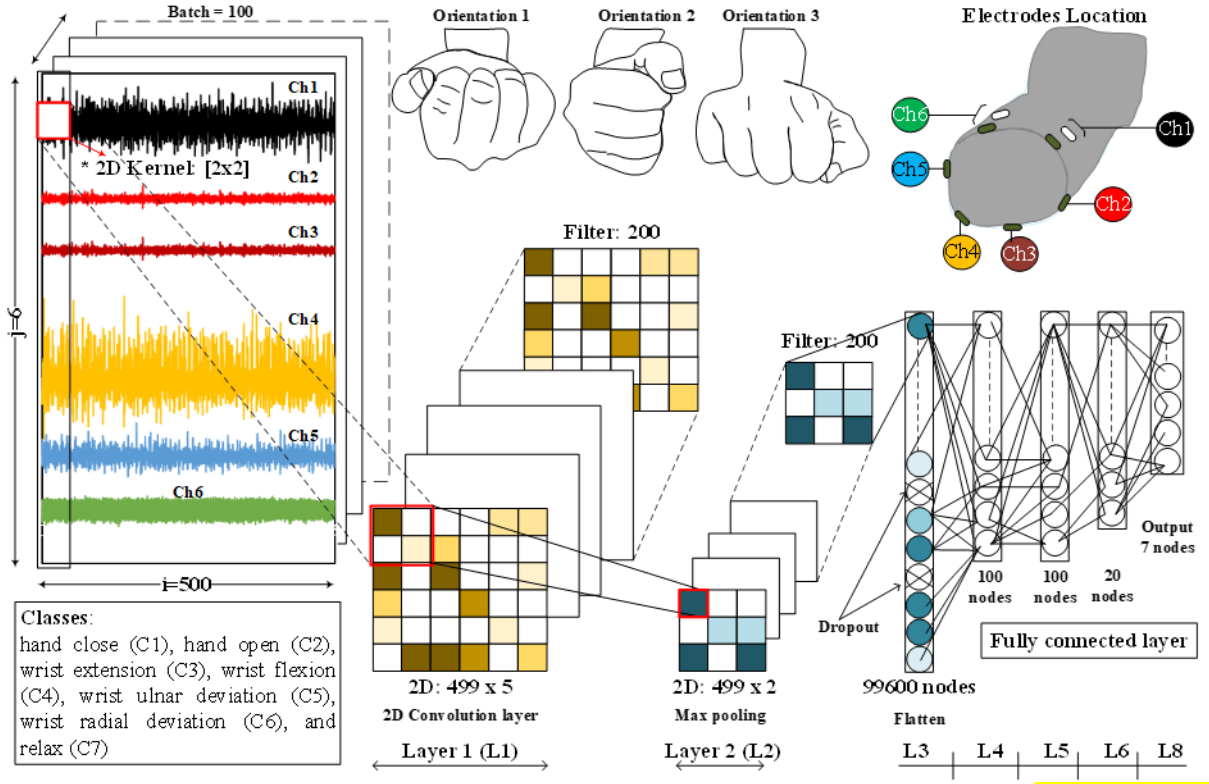


Fig. 1. The implementation of deep learning using two-dimensional convolution layer. (Arial 10, Bold, min 10 words, justify): AUTHOR MUST USE A TEXT BOX FOR IMAGE AND TABLE

relax (C7). The public dataset used in this study is open access and can be found at the following link.

B. Data Collection (H2, Arial 10, BOLD)

(Arial 10, Normal) EMG signals were recorded for approximately 5 seconds or about 20,000 data for every single hand gesture movement. In addition, the subjects were given 10 seconds of rest to prevent the effect of muscle fatigue after each contraction, during data collection. An example of the EMG signal recording dataset is illustrated in Fig. 1. It shows the experiment protocol to collect the EMG signal from one orientation, sequentially. In detail, the number of EMG data for 1 subject and 1 orientation is 1,260,000 (1 orientation x 7 motion x 3 contraction level x 3 trials x 20,000 data). Before being fed to the CNN classifier, the EMG signal was segmented with a window length of 500 samples or 125 milliseconds (fsampling = 4,000 Hz, tsampling = 0.25.

C. Data Processing (H2, Arial 10, Bold)

It can be calculated using Eq. (1) as follows [24]:

$$S(i,j) = (K * I)(i,j) = \sum_m \sum_n I(i-m, j-n) K(m,n) \quad (1)$$

where S (i, j) is a feature map resulting from the convolution between kernel K (m, n) and input feature I (i, j). The output-shape after the 2D convolution process with a 2D kernel can be calculated using Eq. (2), as follows [25] [24]:

$$o = \frac{(w - f + 2p)}{s} + 1 \quad (2)$$

where o, w, f, p, and s are the output shape, input shape, kernel length, and stride, respectively. In the input section, CNN performs a convolution process between 2D kernel and 2D input. This is repeated to produce a features map on the filter.

III. Result (Arial 10, BOLD, H1, Dark Blue)

A. Accuracy (Arial 10, BOLD, H2)

(Arial 10, Normal) The accuracy of all classifiers from both the proposed method (CNN) and the comparison classifier (KNN, SVM, LDA and DT) was calculated for all orientation types (1, 2 and 3). Classifier accuracy was obtained from 10 subjects involved. In this study, the results of the classifier accuracy were displayed using the box plot diagram of descriptive statistics. It is used to visualize the mean and quartile of accuracy values. Fig. 2 shows a comparison of the boxplot accuracy diagrams of the CNN, KNN, SVM, LDA and DT classifiers for all

Manuscript received 8 April 2025; Revised 10 June 2025; Accepted 15 June 2025; Available online 27 June 2025
Digital Object Identifier (**DOI**): https://doi.org/10.35882/jeeemi.v7i3.948
Copyright © 2026 by the authors. This work is an open-access article and licensed under a Creative Commons Attribution-ShareAlike 4.0 International License (CC BY-SA 4.0).

orientation types (1, 2, and 3). The CNN classifier has the highest accuracy (99.3 ± 0.82%, orientation 1) followed by KNN and DT. However, in this case, the SVM classifier produced the worst performance (56.35 ± 10.20%) in the classification process. The comparison of accuracy based on orientation showed that type 1 was the highest for all classifier types followed by orientation types 2 and 3.

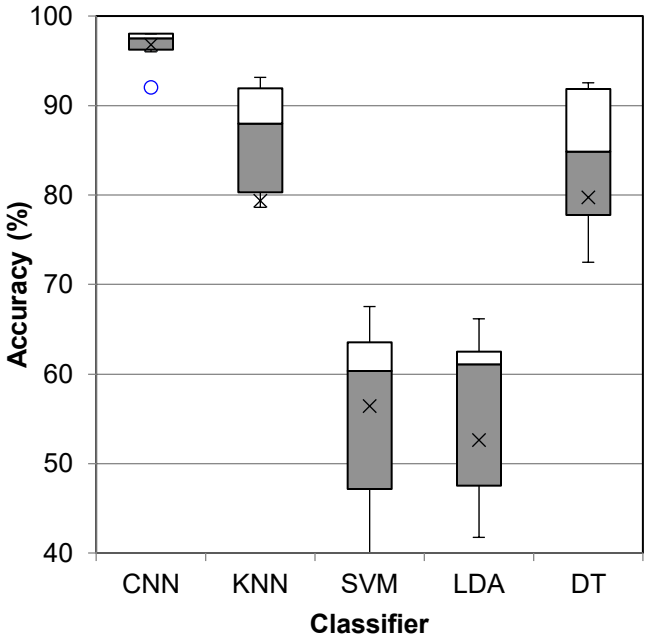


Fig. 2. The boxplot of descriptive statistics to compare the mean accuracy among the classifier for the combination of three orientation and three contraction levels. (Arial 10, Bold, min 10 words, justify).

B. Performance (Arial 10, BOLD)

(Arial 10) In order, the classifier recognized 7 hand gestures for all orientation types and all force levels. Furthermore, three datasets from orientations 1, 2 and 3 were combined and randomized for the training and testing process, which was referred to as scheme 4. Fig. 5 shows that the CNN classifier still produces the best accuracy (96.80 ± 1.87%) compared to other classifiers after the combined process of 3 orientation datasets. Fig. 4 shows a comparison of the boxplot accuracy diagrams of the CNN, KNN, SVM, LDA and DT classifiers for all orientation types (1, 2, and 3). The CNN classifier has the highest accuracy (99.3 ± 0.82%, orientation 1) followed by KNN and DT. However, in this case, the SVM classifier produced the worst performance (56.35 ± 10.20%) in the classification process. The comparison of accuracy based on orientation showed that type 1 was the highest for all classifier types followed by orientation types 2 and 3.

Fig. 4 shows a comparison of the boxplot accuracy diagrams of the CNN, KNN, SVM, LDA and DT classifiers for all orientation types (1, 2, and 3). The CNN classifier has the highest accuracy (99.3 ± 0.82%, orientation 1) followed by KNN and DT. However, in this case, the SVM classifier produced the worst performance (56.35 ± 10.20%) in the classification process. The comparison of accuracy based on orientation showed that type 1 was the highest for all classifier types followed by orientation types 2 and 3. Fig. 4 shows a comparison of the boxplot accuracy diagrams of the CNN, KNN, SVM, LDA and DT classifiers for all orientation types (1, 2, and 3). Table 1 (coloring with dark blue for Table, Figure and Equation) shows the classifier accuracy (mean and standard deviation) for scheme 4. The CNN classifier had the highest accuracy (96.80 ± 1.87%) and lowest standard deviation (± 1.87%) compared to other classifiers.

Table 1 shows the differences and significant different among the accuracy from different orientation. The comparison of the mean accuracy of scheme 4 against schemes 2 and 3 showed that there was no significant difference (4-2 difference: -1.7%, p = 0.121; 4-3 difference: -0.4%, p = 0.949). However, there was a significant difference when the mean accuracy of scheme 4 compared to scheme 1 (4-1 difference: -2.5%, p = 0.010). The comparison of the mean accuracy of scheme 4 against schemes 2 and 3 showed that there was no significant difference (4-2 difference: -1.7%, p = 0.121; 4-3 difference: -0.4%, p = 0.949). However, there was a significant difference when the mean accuracy of scheme 4 compared to scheme 1 (4-1 difference: -2.5%, p = 0.010). The comparison of the mean accuracy of scheme 4 against schemes 2 and 3 showed that there was no significant difference (4-2 difference: -1.7%, p = 0.121; 4-3 difference: -0.4%, p = 0.949). However, there was a significant difference when the mean accuracy of scheme 4 compared to scheme 1 (4-1 difference: -2.5%, p = 0.010).

Table 1. The classifier accuracy (mean and standard deviation) to recognize the seven hand gesture from ten subjects when the data is combined from all type orientation and all contraction. (Arial 10, Bold, min 10 words, justify).

Classifier	Mean (%)	SD (%)
CNN	96.80	±1.87
KNN	79.35	±25.73
SVM	56.36	±10.2

Manuscript received 8 April 2025; Revised 10 June 2025; Accepted 15 June 2025; Available online 27 June 2025
Digital Object Identifier (**DOI**): https://doi.org/10.35882/jeeemi.v7i3.948
Copyright © 2026 by the authors. This work is an open-access article and licensed under a Creative Commons Attribution-ShareAlike 4.0 International License (CC BY-SA 4.0).

LDA	52.62	±17.11
DT	79.67	±18.69

C. Discussion (ARIAL 10, BOLD, H1)

A. Classifier (Arial 10, BOLD, H2)

This study aims to reveal whether there is a significant difference in accuracy when the classifier is tested to recognize 7 gestures using various types of orientation (orientations 1, 2, 3 and all). The results of this study showed that there is a slight decrease in the mean accuracy among scheme 4 against schemes 1, 2 and 3 by (difference 4-1:0.41%, difference 4-2:1.75%, and difference 4-3: 2.58%, respectively).

This indicates that the CNN classifier is able to classify 7 gestures for various types of orientation with low difference in accuracy between orientations (<5%). Furthermore, the single-factor ANOVA statistical test based on Tukey HSD showed that there is no significant difference in accuracy (p-value> 0.05) between the training-testing scheme 4 vs 2, 4 vs 3, 1 vs 2, and 2 vs 3. However, several tests between groups showed that there was a significant difference in accuracy (p-value <0.05) for the scheme group 1 vs 4 and 1 vs 3. This is because, the difference in mean accuracy for the two groups was larger compared to the others (1-4 difference: 2.5% and 1-3 difference: 2.1%). However, the differences are tolerable (<5%). Furthermore, the mean value of accuracy obtained was quite high (96.80 ± 1.87%) when the classifiers were trained and tested using scheme 4. This explains that even though all datasets with three different orientation types were combined, the CNN classifier still showed maximum accuracy. The low standard deviation (1.87%) indicates that most of the resulting accuracy values have almost centered accuracy at 96%. This was different, when accuracy is compared with the standard deviation produced by the KNN, SVM, LDA, and DT with values of ± 25.73%, ± 10.2%, ± 17.11%, and ± 18.69%, respectively.

In addition, the higher accuracy for orientation types 1 and 2 were also in line with the results of study by Rami Khushaba et al. The variations in the accuracy results obtained ranged from 48.6% to 96.6% (mean = 74.8%) for all combinations of training and orientation testing (1, 2 and 3) [11]. Conversely, a decrease in classifier accuracy was also experienced by a study proposed by Yanjuan et al, which examined the effect of the limb position (5 positions) on the classification results. The resulting average accuracy value is ~ 90% [13].

B. Confusion matrices (Heading 2, Arial 10, Bold)

(Arial 10, Normal) The single factor ANOVA statistical test using Tukey HSD on the results of the confusion matrices (Fig. 7) for Post Hoc multiple comparisons showed that there was no significant difference in accuracy (p-value> 0.05) for comparisons between all gestures in the scheme 4. This proved that the proposed classifier (CNN) has the same accuracy for all predicted gestures (7 gestures). In training and testing scheme (4), it is revealed that the resulting accuracy for recognizing gestures (1 = hand open) was the lowest (83.53 ± 10.51%). This low accuracy is because the subject carries out gesture 1, which includes the open position, possibility of each repetition of the open condition is different, causing different EMG records. The low accuracy of gestures (1 = open) is also in line with the results of the study carried out by Khushaba et al, which obtained the lowest accuracy in gestures (1 = hand open) at 84.0% [11].

1. Confusion Matrices CNN (Heading 3, Arial 10)

This proved that the proposed classifier (CNN) has the same accuracy for all predicted gestures (7 gestures). In training and testing scheme (4), it is revealed that the resulting accuracy for recognizing gestures (1 = hand open) was the lowest (83.53 ± 10.51%). This low accuracy is because the subject carries out gesture 1, which includes the open position, possibility of each repetition of the open condition is different, causing different EMG records. The low accuracy of gestures (1 = open) is also in line with the results of the study carried out by Khushaba et al, which obtained the lowest accuracy in gestures (1 = hand open) at 84.0% [11].

2. Confusion Matrice SVM (Heading 3, Arial 10, Norm)

This proved that the proposed classifier (CNN) has the same accuracy for all predicted gestures (7 gestures). In training and testing scheme (4), it is revealed that the resulting accuracy for recognizing gestures (1 = hand open) was the lowest (83.53 ± 10.51%). This low accuracy is because the subject carries out gesture 1, which includes the open position, possibility of each repetition of the open condition is different, causing different EMG records. The low accuracy of gestures (1 = open) is also in line with the results of the study carried out by Khushaba et al, which obtained the lowest accuracy in gestures (1 = hand open) at 84.0% [11].

V. Conclusion (Arial 10, Bold, H1)

(Arial 10, Normal) This study aims to develop a classifier by classifying 7 gestures that are robust against variations of forearm orientation. The result showed that the accuracy of the CNN algorithm outperformed other comparison classifiers (SVM, KNN, LDA, and DT) (p<0.05). There was a decrease in CNN accuracy (<5%) which resulted from the difference from scheme 4

Manuscript received 8 April 2025; Revised 10 June 2025; Accepted 15 June 2025; Available online 27 June 2025
Digital Object Identifier (**DOI**): https://doi.org/10.35882/jeeemi.v7i3.948
Copyright © 2026 by the authors. This work is an open-access article and licensed under a Creative Commons Attribution-ShareAlike 4.0 International License (CC BY-SA 4.0).

(combination of all orientation) to scheme 1, 2, and 3. Furthermore, multiple comparisons using Tukey HSD (alpha = 0.05) revealed that 4 out of 6 groups showed that there was no significant difference in accuracy (p-value> 0.05). The computation time of the proposed CNN was still within the recommended tolerance limit (<200ms). In conclusion, further studies related to the implementation of CNN in embedded systems should be proposed to develop a prosthetic hand that is robust against orientation.

Acknowledgment

The authors would like to express sincere gratitude to the Department of Medical Electronics Technology, Poltekkes Kemenkes Surabaya, for the invaluable support and resources provided throughout this research. The facilities, academic environment, and encouragement from faculty members have significantly contributed to the completion of this work. This study would not have been possible without the institution's commitment to advancing research and innovation in the field of medical electronics.

Funding

This research received no specific grant from any funding agency in the public, commercial, or not-for-profit sectors.

Data Availability

No datasets were generated or analyzed during the current study.

Author Contribution

Triwiyanto conceptualized and designed the study, conducted data collection, and participated in data analysis and interpretation. Bedjo Utomo contributed to the development of the educational media, oversaw the implementation of the intervention, and contributed to manuscript writing and revisions. Sari Luthfiyah assisted with data analysis and interpretation and provided critical feedback on the manuscript. All authors reviewed and approved the final version of the manuscript, and agreed to be responsible for all aspects of the work ensuring integrity and accuracy.

Declarations

Ethical Approval

This study was conducted by ethical standards and has received approval from the Institutional Ethical Board (IRB) of Poltekkes Kemenkes Surabaya, Indonesia, with approval number [045/Polkes/2024]. Informed consent was obtained from the parents or guardians of all participating students, and confidentiality and anonymity of the participants were maintained throughout the research process. All procedures adhered to ethical guidelines for research involving human subjects.

Consent for Publication Participants.

Consent for publication was given by all participants

Competing Interests

The authors declare no competing interests.

References (Heading 1, Arial 10, Bold, Blue)

[1] M. Asghari Oskoei and H. Hu, “Myoelectric control systems—A survey,” *Biomed. Signal Process. Control*, vol. 2, no. 4, pp. 275–294, Oct. 2007. (Arial 10)

[2] C. Fleischer and A. Wege, “Application of EMG signals for controlling exoskeleton robots EMG is better than force sensor,” *Biomed Tech*, vol. 51, pp. 314–319, 2006.

[3] J. V. V Basmajian and C. J. De Luca, “Chapter2: Apparatus, Detection, and Recording,” *Muscles alive their Funct. Reveal. by Electromyogr.*, vol. 2, pp. 19–65, 1985.

[4] J. V Basmajian and C. J. De Luca, “Chapter1: introduction,” *Muscles Alive their Funct. Reveal. by Electromyogr.*, pp. 1–18, 1985.

[5] G. R. Naik, S. Member, A. Al-timemy, and H. T. Nguyen, “Transradial Amputee Gesture Classification using an Optimal Number of sEMG Sensors : An Approach using ICA Clustering,” *IEEE Trans. Neural Syst. Rehabil. Eng.*, vol. 4320, no. c, 2015.

[6] T. Triwiyanto, I. P. A. Pawana, and M. H. Purnomo, “An Improved Performance of Deep Learning Based on Convolution Neural Network to Classify the Hand Motion by Evaluating Hyper Parameter,” *IEEE Trans. Neural Syst. Rehabil. Eng.*, vol. 28, no. 7, pp. 1678–1688, 2020.

[7] M. Atzori, M. Cognolato, and H. Müller, “Deep Learning with Convolutional Neural Networks Applied to Electromyography Data : A Resource for the Classification of Movements for Prosthetic Hands,” *Front. Neurorobot.*, vol. 10, no. September, pp. 1–10, 2016.

[8] U. Côté-Allard, C. L. Fall, A. Campeau-Lecoursy, C. Gosseliny, F. Laviolettez, and B. Gosselin, “Transfer learning for sEMG hand gestures recognition using convolutional neural networks,” *2017 IEEE Int. Conf. Syst. Man, Cybern. SMC 2017*, vol. 2017–Janua, pp. 1663–1668, 2017.

[9] T. Triwiyanto, O. Wahyunggoro, H. A. Nugroho, and H. Herianto, “Muscle fatigue compensation of the electromyography signal for elbow joint angle estimation using adaptive feature,” *Comput. Electr. Eng.*, vol. 71, no. July, pp. 284–

Manuscript received 8 April 2025; Revised 10 June 2025; Accepted 15 June 2025; Available online 27 June 2025
Digital Object Identifier (**DOI**): https://doi.org/10.35882/jeeemi.v7i3.948
Copyright © 2026 by the authors. This work is an open-access article and licensed under a Creative Commons Attribution-ShareAlike 4.0 International License (CC BY-SA 4.0).

293, Oct. 2018.

[10] A. H. Al-Timemy, R. N. Khushaba, G. Bugmann, and J. Escudero, "Improving the Performance Against Force Variation of EMG Controlled Multifunctional Upper-Limb Prostheses for Transradial Amputees," *IEEE Trans. Neural Systdems Rehabil. Eng.*, vol. 24, no. 6, pp. 650–661, Jun. 2016.

[11] R. N. Khushaba, A. Al-timemy, S. Kodagoda, and K. Nazarpour, "Combined influence of forearm orientation and muscular contraction on EMG pattern recognition," *Expert Syst. Appl.*, vol. 61, pp. 154–161, 2016.

[12] E. Scheme, A. Fougner, Ø. Stavdahl, and A. D. C. Chan, "Examining the Adverse Effects of Limb Position on Pattern Recognition Based Myoelectric Control," in *32nd Annual International Conference of the IEEE EMBS*, 2010, pp. 6337–6340.

[13] Y. Geng, P. Zhou, and G. Li, "Toward attenuating the impact of arm positions on electromyography pattern-recognition based motion classification in transradial amputees," *J. Neuroeng. Rehabil.*, vol. 9, no. 1, p. 1, 2012.

[14] D. Yang, W. Yang, Q. Huang, H. Liu, and S. Member, "Classification of Multiple Finger Motions during Dynamic Upper Limb Movements," *IEEE J. Biomed. Heal. Informatics*, vol. 21, no. 1, pp. 134–141, 2015.

[15] J. L. Betthauser, C. L. Hunt, L. E. Osborn, R. R. Kaliki, and N. V Thakor, "Limb-Position Robust Classification of Myoelectric Signals for Prosthesis Control Using Sparse Representations," in *Annu Int Conf IEEE Eng Med Biol Soc*, 2016, pp. 6373–6376.

[16] L. Chen, G. Yanjuan, and L. Guanglin, "Effect of Upper-Limb Positions on Motion Pattern Recognition Using Electromyography," in *2011 4th International Congress on Image and Signal Processing*, 2011, pp. 139–142.

[17] Triwiyanto, O. Wahyunggoro, H. A. Nugroho, and Herianto, "An investigation into time domain features of surface electromyography to estimate the elbow joint angle," *Adv. Electr. Electron. Eng.*, vol. 15, no. 3, pp. 448–458, 2017.

[18] A. Phinyomark, P. Phukpattaranont, and C. Limsakul, "Feature reduction and selection for EMG signal classification," *Expert Syst. Appl.*, vol. 39, no. 8, pp. 7420–7431, 2012.

[19] R. N. Khushaba, S. Member, A. H. Al-timemy, A. Al-ani, and A. Al-jumaily, "A Framework of Temporal - Spatial Descriptors based Feature Extraction for Improved Myoelectric Pattern Recognition," *IEEE Trans. Neural Syst. Rehabil. Eng.*, vol. 4320, no. c, 2017.

[20] P. Geethanjali and K. K. Ray, "A Low-Cost Real-Time Research Platform for EMG Pattern Recognition-Based Prosthetic Hand," *IEEE/ASME Trans. Mechatronics*, vol. 20, no. 4, pp. 1948–1955, Aug. 2015.

[21] A. Phinyomark and E. Scheme, "EMG Pattern Recognition in the Era of Big Data and Deep Learning," *Big Data Cogn. Comput.*, vol. 2, no. 3, p. 21, Aug. 2018.

[22] A. Hartwell, V. Kadirkamanathan, and S. R. Anderson, "Compact Deep Neural Networks for Computationally Efficient Gesture Classification From Electromyography Signals," *arXiv:1806.08641v2 [cs.CV]*, Jun. 2018.

[23] A. Sors, S. Bonnet, S. Mirek, L. Vercueil, and J.-F. Payen, "A convolutional neural network for sleep stage scoring from raw single-channel EEG," *Biomed. Signal Process. Control*, vol. 42, pp. 107–114, Apr. 2018.

[24] C. Ieracitano, N. Mammone, A. Bramanti, A. Hussain, and F. C. Morabito, "A Convolutional Neural Network approach for classification of dementia stages based on 2D-spectral representation of EEG recordings," *Neurocomputing*, vol. 323, pp. 96–107, 2019.

[25] T. Epelbaum, "Deep learning: Technical introduction," in *arXiv:1709.01412*, arxiv.org, 2017, pp. 1–106.

[26] L. H. Smith, L. J. Hargrove, B. a. Lock, and T. a. Kuiken, "Determining the optimal window length for pattern recognition-based myoelectric control: Balancing the competing effects of classification error and controller delay," *IEEE Trans. Neural Syst. Rehabil. Eng.*, vol. 19, no. 2, pp. 186–192, 2011.

Author Biography (ARIAL 10, BOLD, H1)

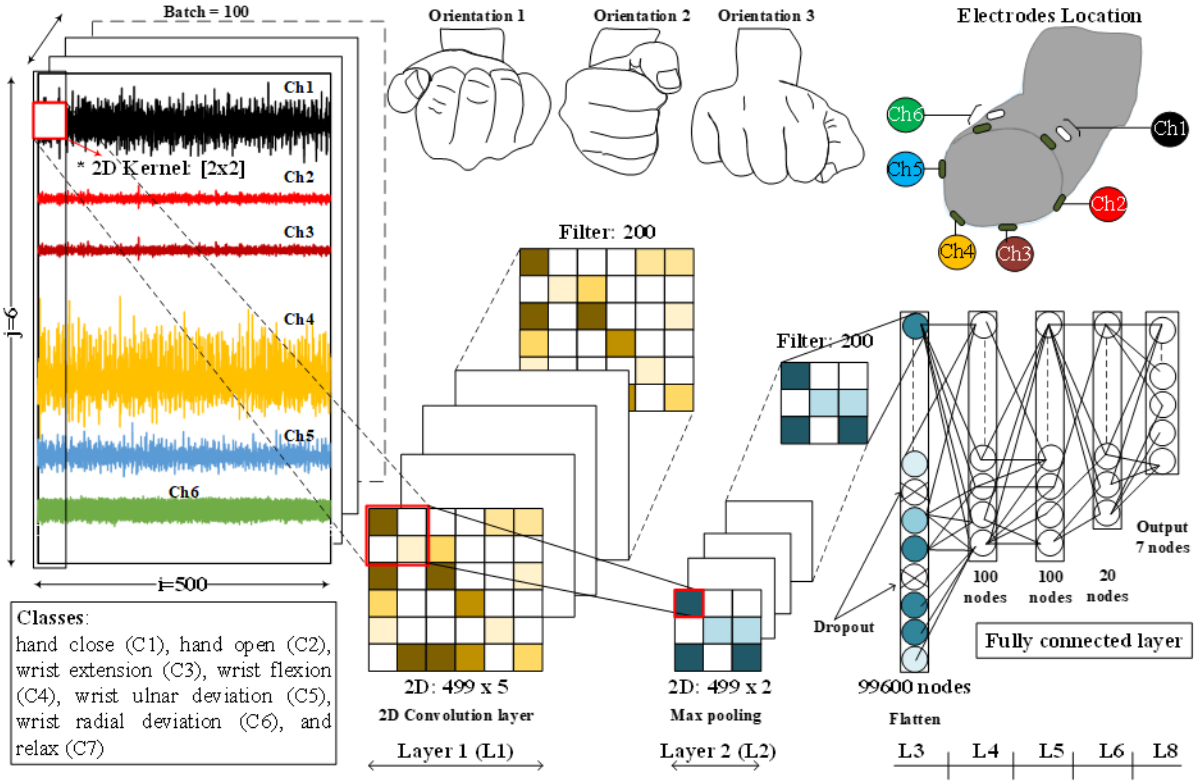


Triwiyanto Triwiyanto (S' 2015 - M '2024) received the B.S. degree in Physics (Instrumentation) from Airlangga University, in 1997, M. Eng. degrees in Electronic Engineering from the Institut Teknologi Sepuluh Nopember Surabaya, Indonesia in 2004, and the Ph.D. degree in Electrical Engineering from Gadjah Mada University,Yogyakarta, Indonesia, in 2018. Since 2005, he has been an Associate Professor with the Medical Electronics Technology, Health Polytechnic Ministry of Health Surabaya, Indonesia. Since 2015, he is an IEEE

Manuscript received 8 April 2025; Revised 10 June 2025; Accepted 15 June 2025; Available online 27 June 2025
Digital Object Identifier (**DOI**): https://doi.org/10.35882/jeeemi.v7i3.948
Copyright © 2026 by the authors. This work is an open-access article and licensed under a Creative Commons Attribution-ShareAlike 4.0 International License (CC BY-SA 4.0).

member. His current research interests include biomedical signal processing, rehabilitation engineering, machine learning, and surface electromyography (sEMG)-based physical human robot interactions. Currently, He has more than 100 documents on Scopus database (Arial 10)

Each author is required to provide a cover image as shown example below





Wahyu Caesarendra is an Assistant Professor at the Faculty of Integrated Technologies, Universiti Brunei Darussalam since October 2018. He received a Bachelor of Engineering degree from Diponegoro University, Indonesia in 2005. He worked in the Department of Mechanical Engineering, Diponegoro University from 2010 to 2018 as an Assistant Professor. He received New University for Regional Innovation (NURI) and Brain Korea 21 (BK21) scholarships for Master study in 2008 and obtained his Master of Engineering (M.Eng) degree from Pukyong National University, South Korea in 2010. In 2011, Wahyu Caesarendra was awarded of University Postgraduate Award (UPA) and International Postgraduate Tuition Award (IPTA) from the University of Wollongong. He received a Doctor of Philosophy (Ph.D.) Degree from the University of Wollongong in 2015.



Mauridhi Hery Purnomo (IEEE Senior Member) received the B. S degree in power system engineering from Institut Teknologi Sepuluh Nopember Surabaya, Indonesia, in 1985. M. Eng and Ph.D degrees in Control Engineering and Intelligence System from Osaka City University Japan, in 1995 and 1998. He is a professor of Computer Engineering Department of Sepuluh Nopember Institute of Technology, Surabaya, Indonesia. He is an IEEE senior member. His research interests include control systems, artificial intelligent, application of neural networks and deep learning application for power system.

Manuscript received 8 April 2025; Revised 10 June 2025; Accepted 15 June 2025; Available online 27 June 2025
Digital Object Identifier (**DOI**): https://doi.org/10.35882/jeeemi.v7i3.948
Copyright © 2026 by the authors. This work is an open-access article and licensed under a Creative Commons Attribution-ShareAlike 4.0 International License (CC BY-SA 4.0).